Adani University

Bachelor of Technology

Subject Name: Data Communication & Computer Networks

Subject Code: EICCI24203 Sem: 4th Date: 23-02-2026

URN. No. 2410111124 Name: Sujal Patel Sec : B

EXPERIMENT: 8

OBJECTIVE:

To design and stimulate a Metropolitan Area Network (MAN) using PCs, Switches and Routers.

SOFTWARE USED:

CISCO PACKAGE TRACER, DESKTOP

THEORY:

Advantages

- Connects multiple LANs across a city or metropolitan region.
- Provides high-speed data communication using fiber optic or broadband links.
- Allows efficient sharing of resources like servers, databases, and internet connections.
- Supports a large number of users and handles heavy network traffic effectively.

Disadvantages

- High setup and infrastructure cost.
- Requires skilled professionals for maintenance and management.
- Increased security risks if proper protection measures are not implemented.

CODE:

Not applicable

PRACTICAL:

1.With 6 computers:

Figure 1A :- Basic Diagram

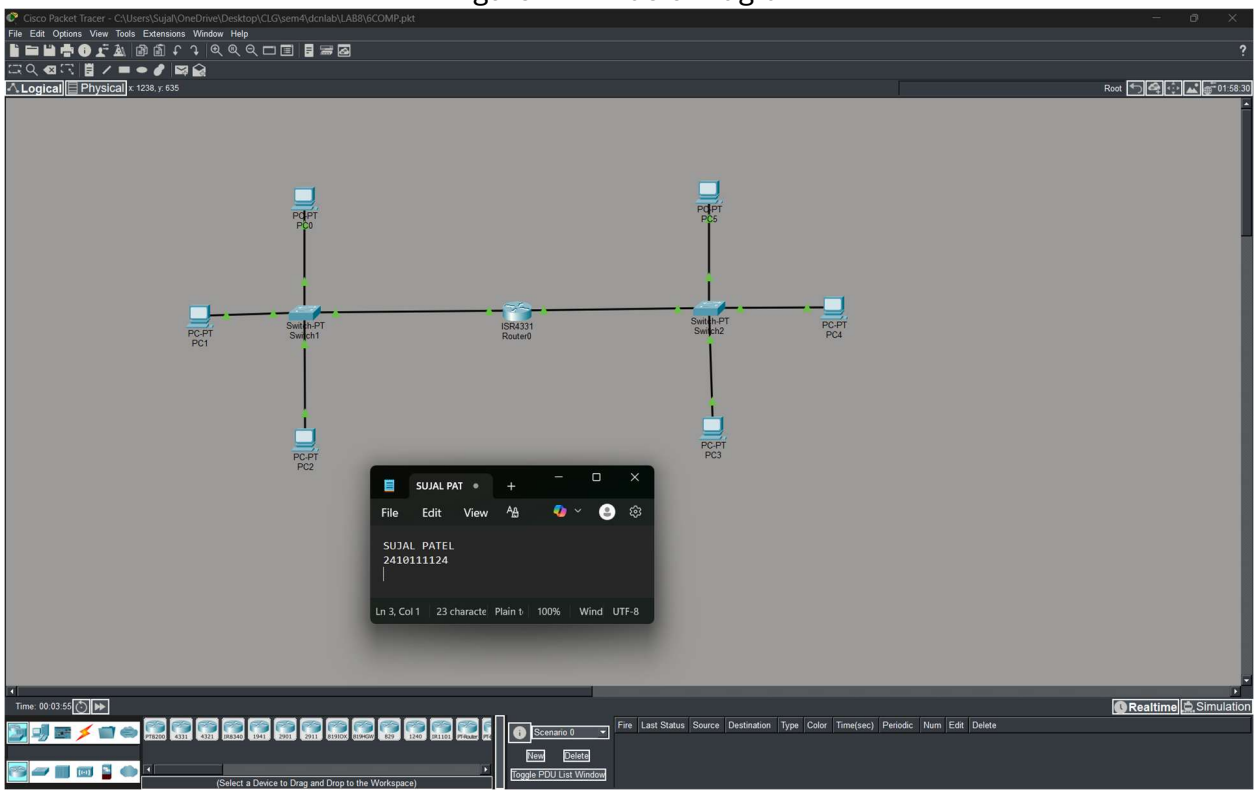


Figure 1B :- Simulation Diagram

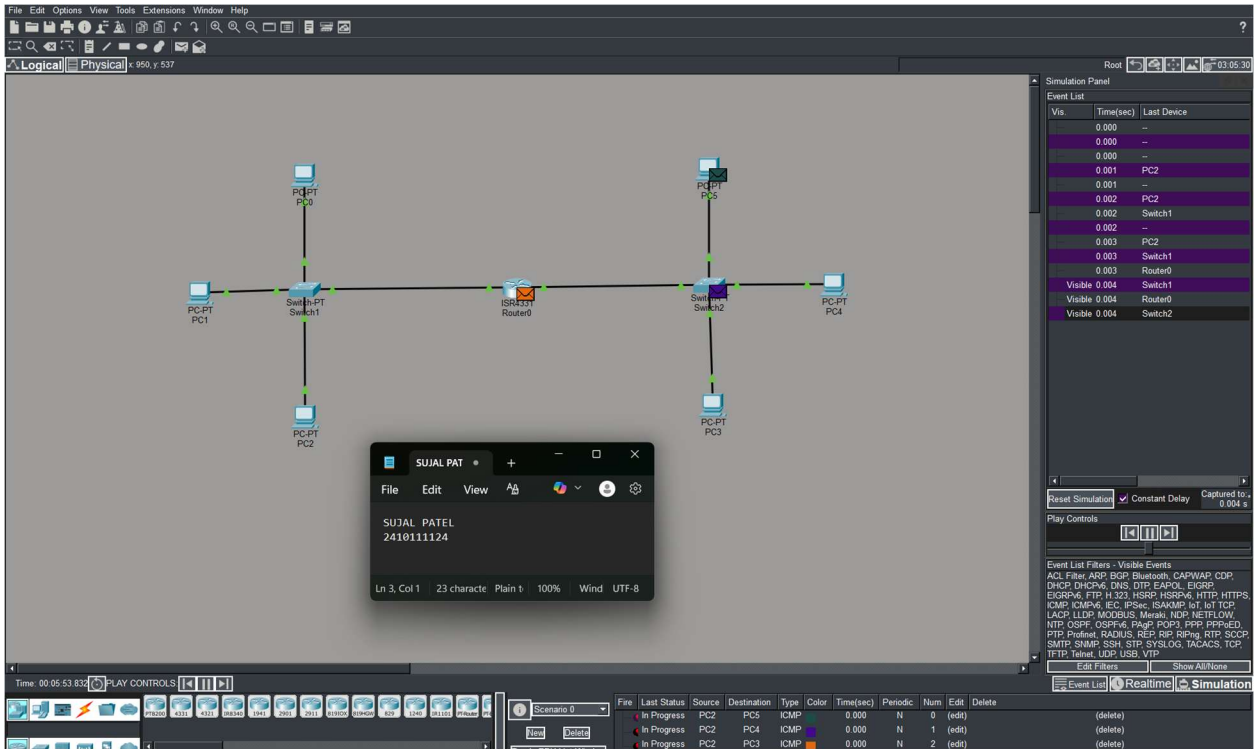
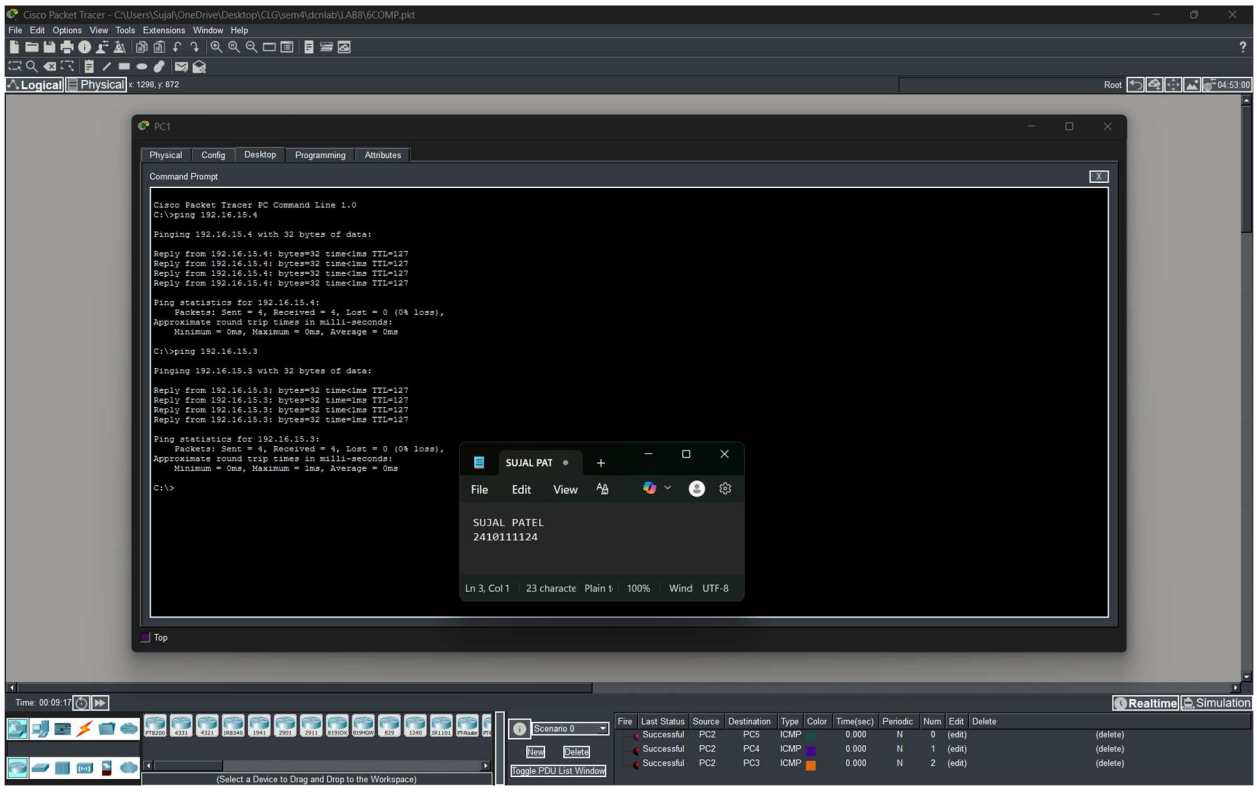


Figure 1C :- Analysis of message received



2.With 14 Computers :

Figure 2A :- Basic Diagram

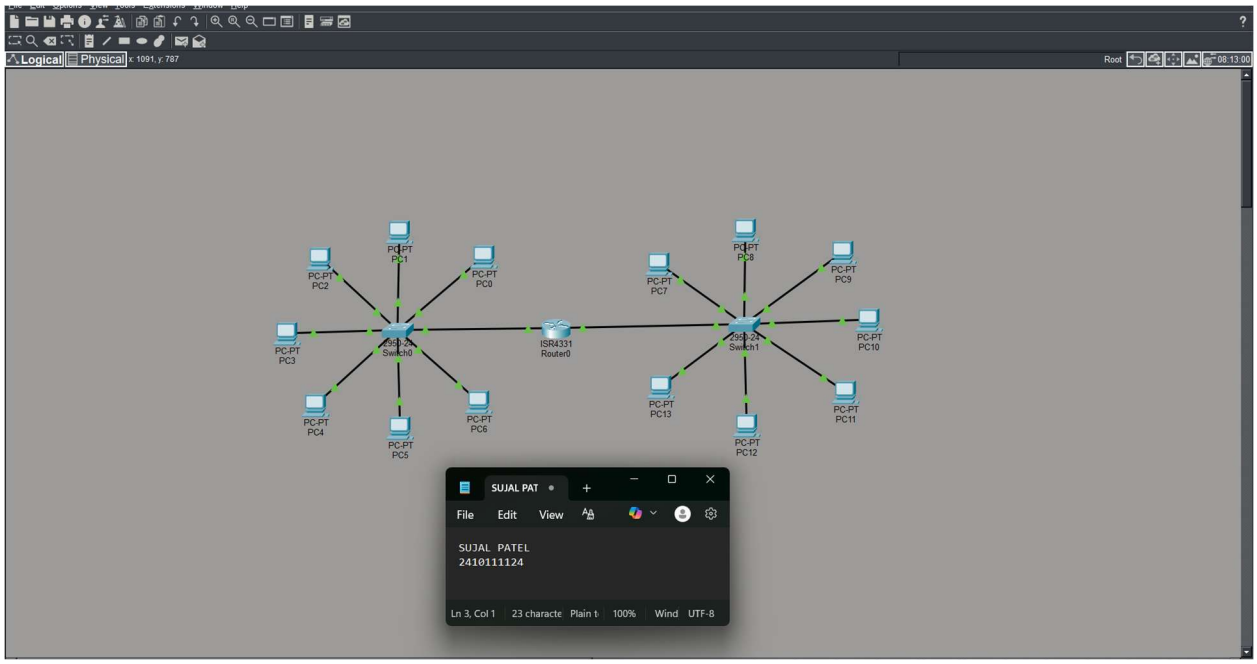


Figure 2B :- Simulation Diagram

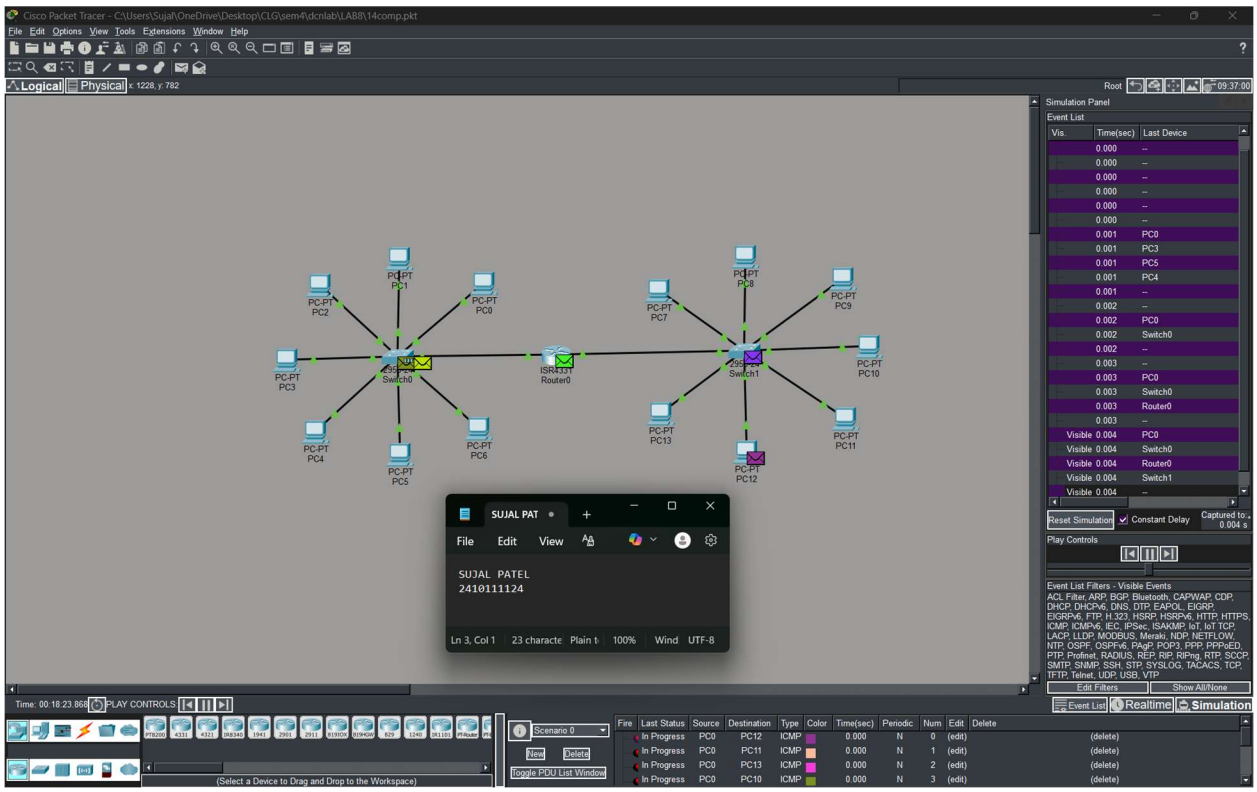
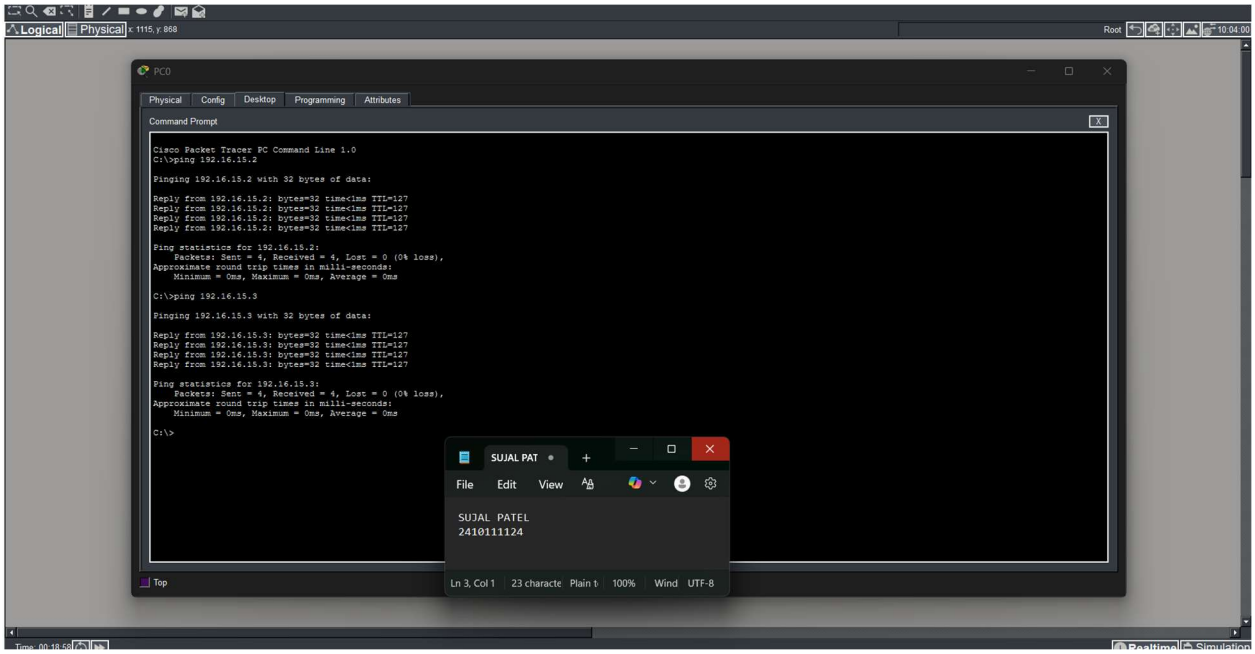
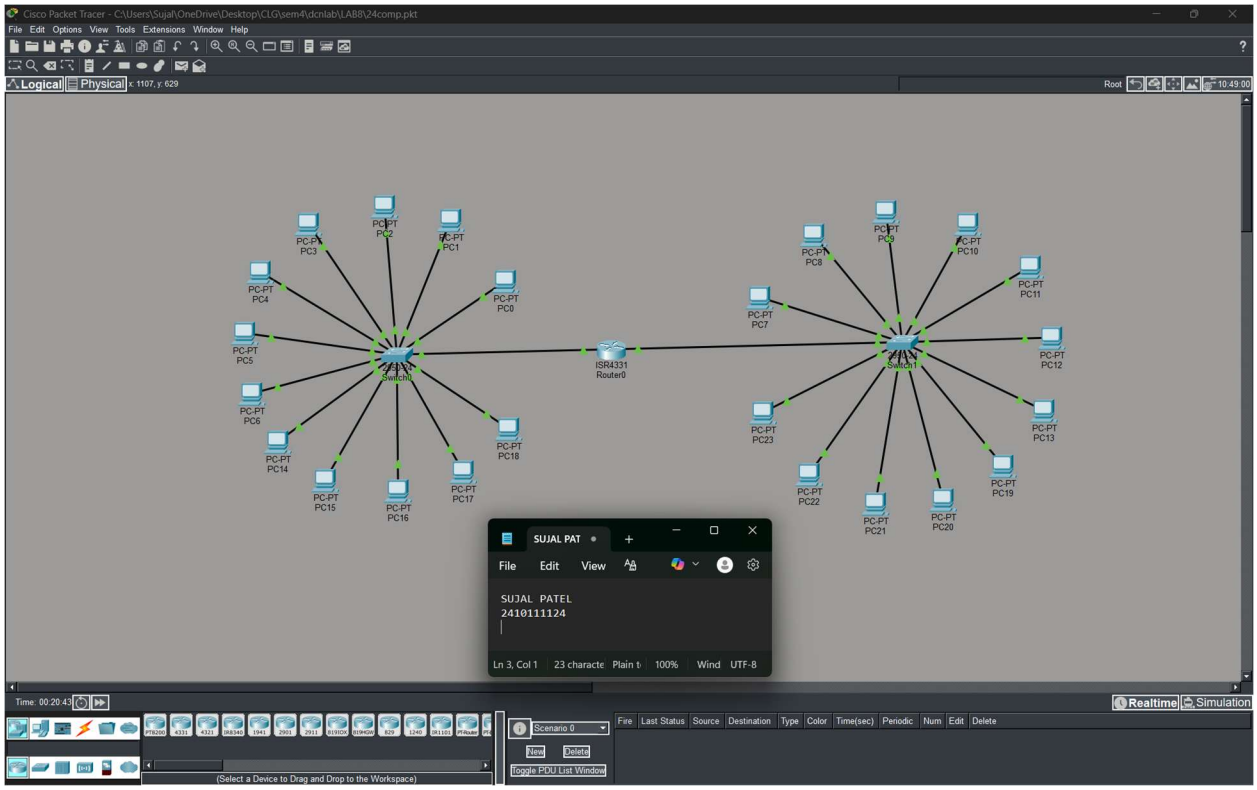


Figure 2C :- Analysis of message received

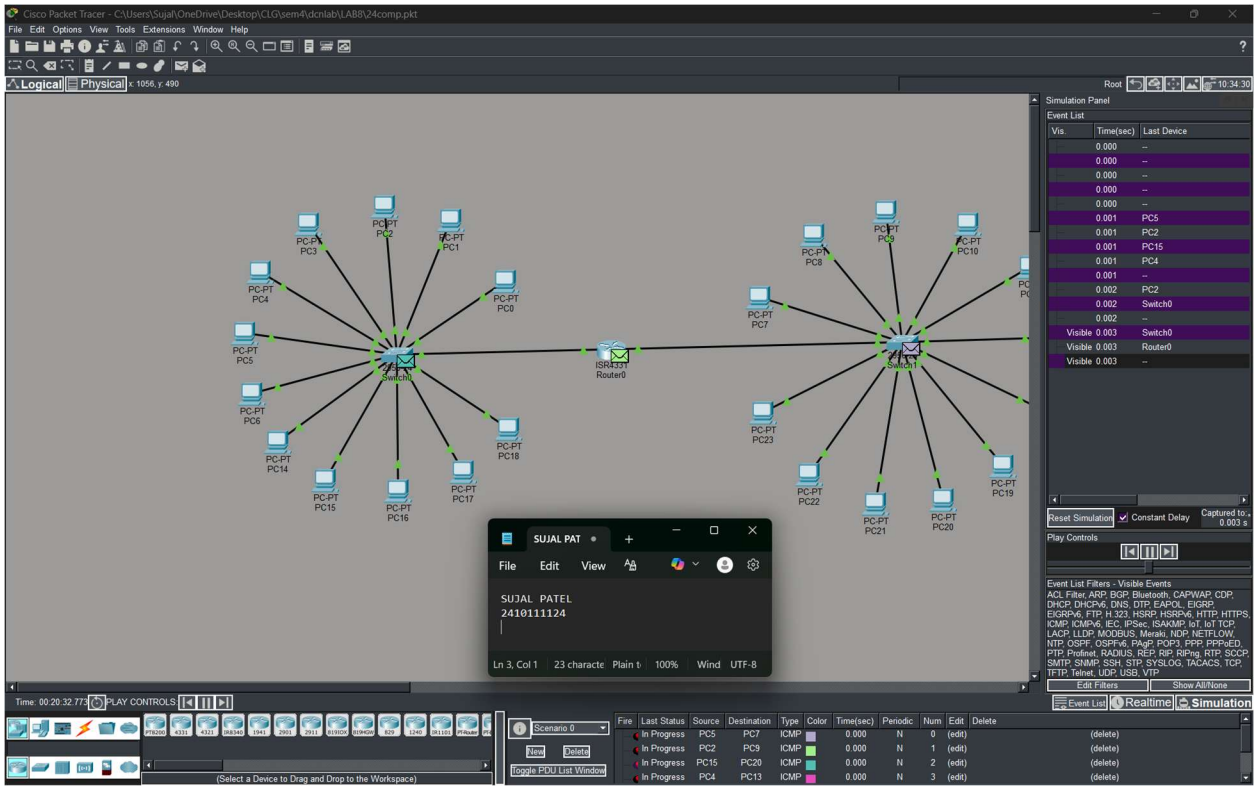


3.With 24 Computers

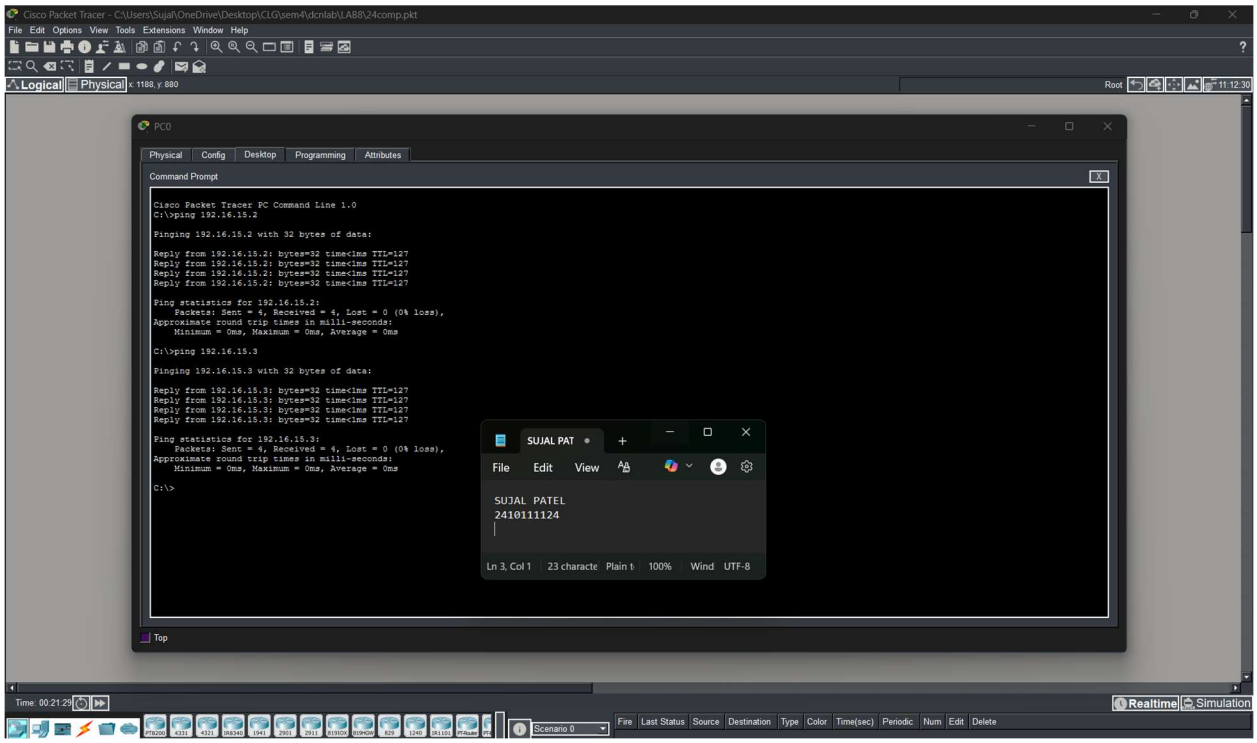
Level 3A :- Basic Diagram



Level 3B : - Simulation Diagram



Level 3C :- Analysis of message received



PROCEDURE:

1. Cisco Packet Tracer was launched and a new project was created.
2. The necessary number of PCs, switches and server were placed in the workspace.
3. PCs were grouped and connected to a switch in a standard topology.
4. A switch connected to the server for establishing communication.
5. Static IP addresses and DNS address within the same network were assigned to all PCs.
6. Static IP address is configured on the server.
7. The network was allowed to stabilize in real-time mode.
8. Message transmission was verified using simulation mode.

RESULTS:

The implementation of the MAN was verified successfully. Multiple LANs are connected to the router for simulation of a city wide network. End devices were able to communicate with devices from the other network by proper routing configuration. Communication between different network segments was successfully established through accurate IP addressing and proper configuration of router interfaces. Each network of devices is given different gateway addresses.

POSSIBLE SOURCES OF ERROR:

1. Incorrect IP address or subnet mask configuration.
2. Entering the IP address in the default gateway field instead of the IPv4 address field.
3. Improper or loose cable connections between devices.
4. Testing the network before the network devices were fully initialized and ready.

CONCLUSION:

The simulation of a Metropolitan Area Network (MAN) was successfully completed using Cisco Packet Tracer. By interconnecting multiple LANs through routers and WAN links, a large-scale network covering a metropolitan region was established. This experiment highlighted the importance of routing, proper IP addressing, and inter-network communication in building scalable and efficient network infrastructures. The MAN topology demonstrated how geographically separated networks can function as a unified system.

LEARNING OUTCOME:

1. Understand the concept of a Metropolitan Area Network (MAN**)** and how it connects multiple LANs within a city.
2. Understood the role of routers in connecting geographically separated networks.
3. Assigned and managed IP addressing across different network segments.

PRECAUTIONS:

1. Ensure each PC has a unique IP address within the same subnet.
2. Use appropriate cable types for device connections.
3. Leave the default gateway empty when no router is used.
4. Allow sufficient time for the network to stabilize before testing communication.